Week 7 · Screens, and judging them: wireframing & usability

Technical Interaction Design · ITU · Autumn 2026

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Usability

Source: Lectures/Interaction-Design/Usability-and-Its-Evaluation.md — something unclear? Open an issue, or send a pull request.

Usability is a quality attribute that assesses how easy user interfaces are to use. It is usually thought as decomposable further in more precise qualities:

1. **Learnability** - How easy is it for users to accomplish basic tasks the first time they encounter the design?
2. **Efficiency** - Once users have learned the design, how quickly can they perform tasks?
3. **Memorability** - When users return to the design after a period of not using it, how easily can they reestablish proficiency?
4. **Absence of Errors** - How many errors do users make, how severe are these errors, and how easily can they recover from the errors?
5. **Satisfaction** - How pleasant is the use of the design?

Heuristics

Heuristics – loosely defined rules that are likely to help, but are not laws - they can be “broken” if necessary.

One of the most popular set of heuristics for interactive systems is the one of Jakob Nielsen.

Published in 1994 in the book “Usability Inspection Methods” by Jakob Nielsen and Robert L. Mack. They were a refinement of an earlier set of heuristics by Nielsen and Molich – a dane.

1. Visibility of system status
2. Match between system and the real world
3. User control and freedom
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5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use
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9. Help users recognize, diagnose, and recover from errors
10. Help and documentation

Heuristic evaluation is based *solely* on an agreed set of recognized heuristics

Other sets of heuristics exist besides the Nielsen ones. E.g., ISO standard 9241-110 from 2020 proposes:

- seven principles,
- 20 categories of recommendations,

- 65 recommendations, and about
- 140 examples take up 15 densely printed pages in the standard

Limitation of heuristic evaluation: usability is more complex than what can be captured in 10 rules.

For you: Learn one set of heuristics. An expert *has the knowledge in their head*.

Usability Testing

The golden standard of usability evaluation is usability testing. This is when you evaluate with users a prototype or an actual system.

Rules of thumb:

- The more detailed and realistic your prototype, the more you can trust your results; the more effort it takes to make changes if you've made a wrong decision
- Even with less detailed prototypes, you can detect many problems
- Don't use expensive testing to find out things you can detect with cheap tests (Hall, 2013)

Sometimes you don't start from scratch: evaluating an existing prototype for usability is even easier - you don't have to create the prototype. You go straight to the next steps. Sometimes, you might even learn useful stuff from usability testing the competitor's products (Hall, 2013).

The Participants

Three kinds of participants

- The log keeper - writes down observations
- The facilitator - explains and helps cautiously, when needed
- The users - test the prototype / system

The Users

Must be representative for your target population.

Best is to have them think aloud. You can also just observe, but it's less valuable.

After your first mockup, probably one or two users are sufficient – if they are representative. You'll get sufficient feedback to go back to the drawing board.

The Tasks

You need to prepare a **set of tasks** that are representative or important for the system. Then you ask the users to try to solve them using the system.

The tasks should be expressed from the POV of the user, not from the POV of the system. They should express a need or a problem that a typical user of the system encounters.

Avoid suggesting which UI elements to be used. That will skew the results.

Avoid trivial tasks. The tasks should be “something that should give the user a feeling of achievement when they are done”. As Lauesen proposes, a task is something after which the user can tell himself: I've done that, I can go and have a coffee now. What does this mean about logging in as a task?

Test Report

The report ranks and prioritizes the problems discovered during the usability testing. It is different than the log. It summarizes the findings based on all the participants.

One possible way of classifying the tasks severity is proposed by Lauesen who argues for five severity classes:

- Missing functionality or bug
- Task failure
- Cumbersome
- Medium problem: succeeds, but user is slow
- Minor: success with a bit of hesitation

Another, simpler classification is by Erika Hall (2013): she argues that three classes that are defined based on a combination of frequency and impact suffices:

- High-impact problems that prevent the user from completing a task. If you don't resolve these you have a high risk to the success of your product.
- Moderate problems with low frequency or low problems with moderate frequency
- Low-impact problems affecting a low number of users. Obviously these don't get prioritized.

Bibliography

- *User Interface Design*, Chapter 13: *More on usability testing*. Søren Lauesen, Addison Wesley.
- *Just Enough Research*, Chapter 7: *Evaluative Research*, Erika Hall, 2013.
- *Nielsen Heuristics with Examples*, Rolf Molich. Web Page at Rolf Molich's Websitej
- *Turn User Goals into Task Scenarios for Usability Testing*, Marieke McCloskey Web Page at nngroup
- The Design of Everyday Things , Don Norman.

Project Work

- Evaluate the usability of your Lo-Fi prototype
- Record brief notes of the usability test outcome
- Write a usability report in which you prioritize the discovered issues
- Improve design

Visibility of system status

Source: Lectures/Interaction-Design/usability-examples/1_status.md — something unclear? Open an issue, or send a pull request.

The system should always keep users informed about what is going on, through appropriate feedback within reasonable time.

Reporting progress heuristic: if waiting time is longer than 0.5s, show progress feedback. If longer than 5s show time estimation if possible.

Counterexamples

Status Impossible to Guess

Miele Dishwasher at ITU requires an analog sticker that employees are supposed to update to show the state!

It is not clear when it is running or not, even if there is a digital display! In fact, when first time users are looking at the display it shows a time, and a temperature, it makes it seem like the dishwasher is actually running. But it is not - it simply shows the details of the currently selected programme.

This is why somebody had to stick an external status gear, that now users have to manually update. This is design against humanity.

Something unclear in this section? Open an issue.

Match between system and the real world

Source: Lectures/Interaction-Design/usability-examples/2_match.md — something unclear? Open an issue, or send a pull request.

The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system-oriented terms.

Follow real-world conventions, making information appear in a natural and logical order.

Counterexamples

Twitter And the :heart Emoji

Mircea: I want to show support for Crista. But the little "heart" emoji is not the best way. In the real world I would express sadness, or concern. The fact that on x-twitter you have to "love" everything shows a mismatch between the system and the real world. It also shows a lack of 4_consistency with the rest of similar applications with which we are .

Something unclear in this section? Open an issue.

User control and freedom

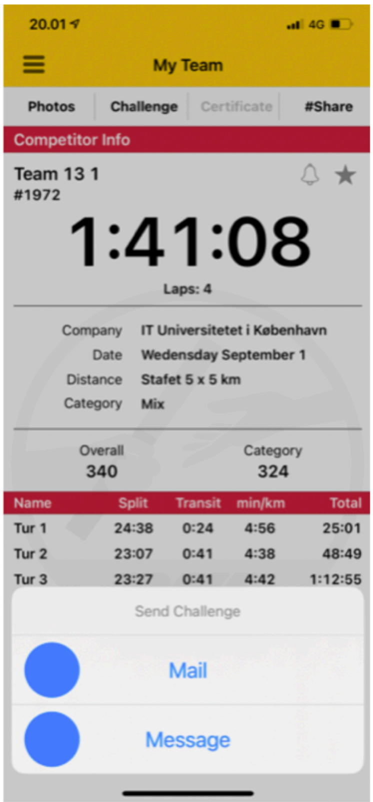
Source: Lectures/Interaction-Design/usability-examples/3_control.md — something unclear? Open an issue, or send a pull request.

Users often choose system functions by mistake and will need a clearly marked “emergency exit” to leave the unwanted state without having to go through an extended dialogue. Support undo and redo.

Counterexamples

No Way Back

The DHL app does not allow me to go back after clicking on Send a Challenge by mistake. I have to kill the app and start again ^^!



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Consistency and standards

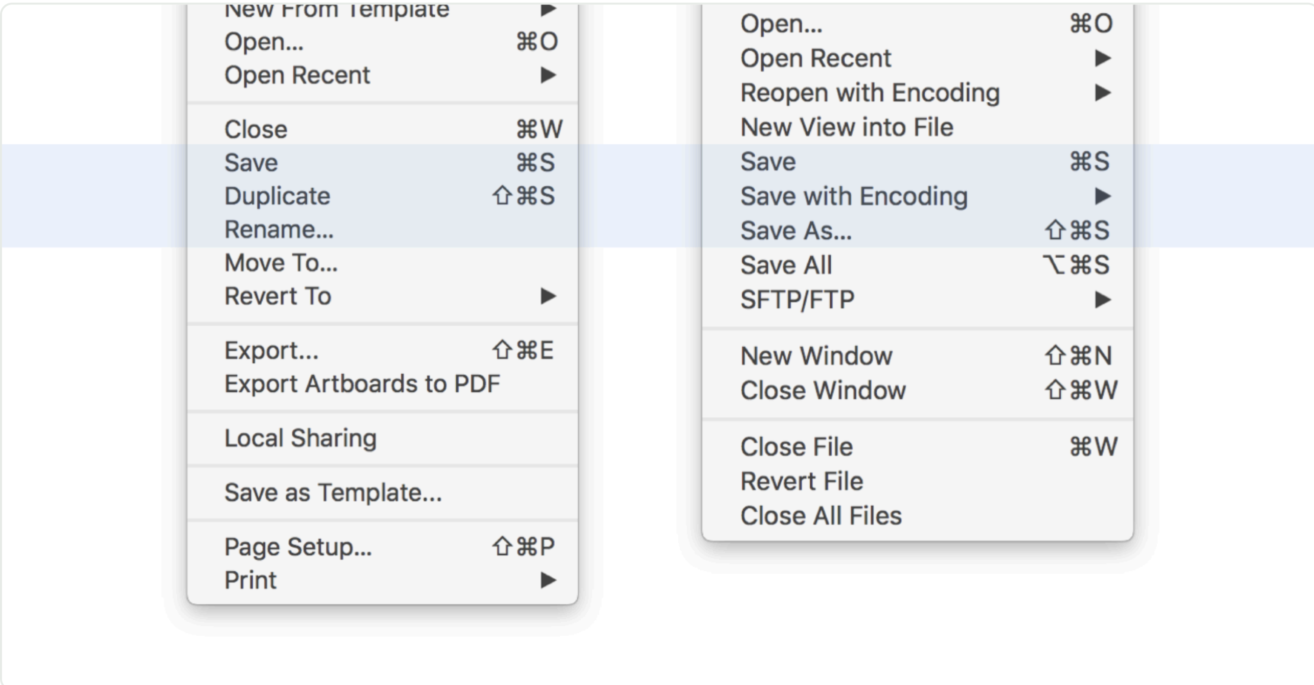
Source: Lectures/Interaction-Design/usability-examples/4_consistency.md — something unclear? Open an issue, or send a pull request.

Consistency and standards means that words, situations, actions and affordances should mean the same thing across different elements within your app, and across multiple platforms.

Examples

Apple's Duplicate Menu Option

E.g. All the world used to know what *Save* and *Save As ..* meant. Then Apple introduced the *Duplicate* + *Rename* idea as a replacement for *Save As...* people hated it. Besides being inconsistent with the rest of the UI world, the pattern also is a failure of another heuristic: 7_flexibility_and_efficiency.



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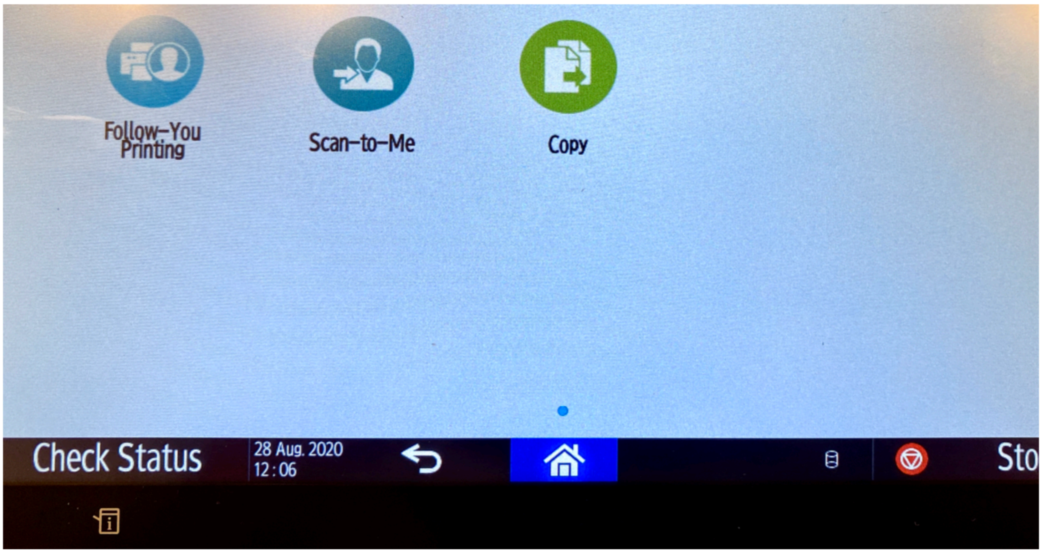
Counterexamples

Inconsistent Terminology

Mircea: The three options on the screen of some of the printers at ITU (as of 2023, and since 2020 at least) have frustrating labels. Follow-**You**, Scan-to-**Me** ... Who is You? Who is Me? Are they the same person? Probably. Then why not unify the options. The fact that Copy does not have a pronoun seems also quite inconsistent.



Two Buttons - Same Action

Vibeke: The add date read pop-up box on the goodreads.com website has two options of stopping the action happening, there is no indication of the difference between close and cancel. This creates a illusion that there is a difference between the two options, when in fact they result in the same action.

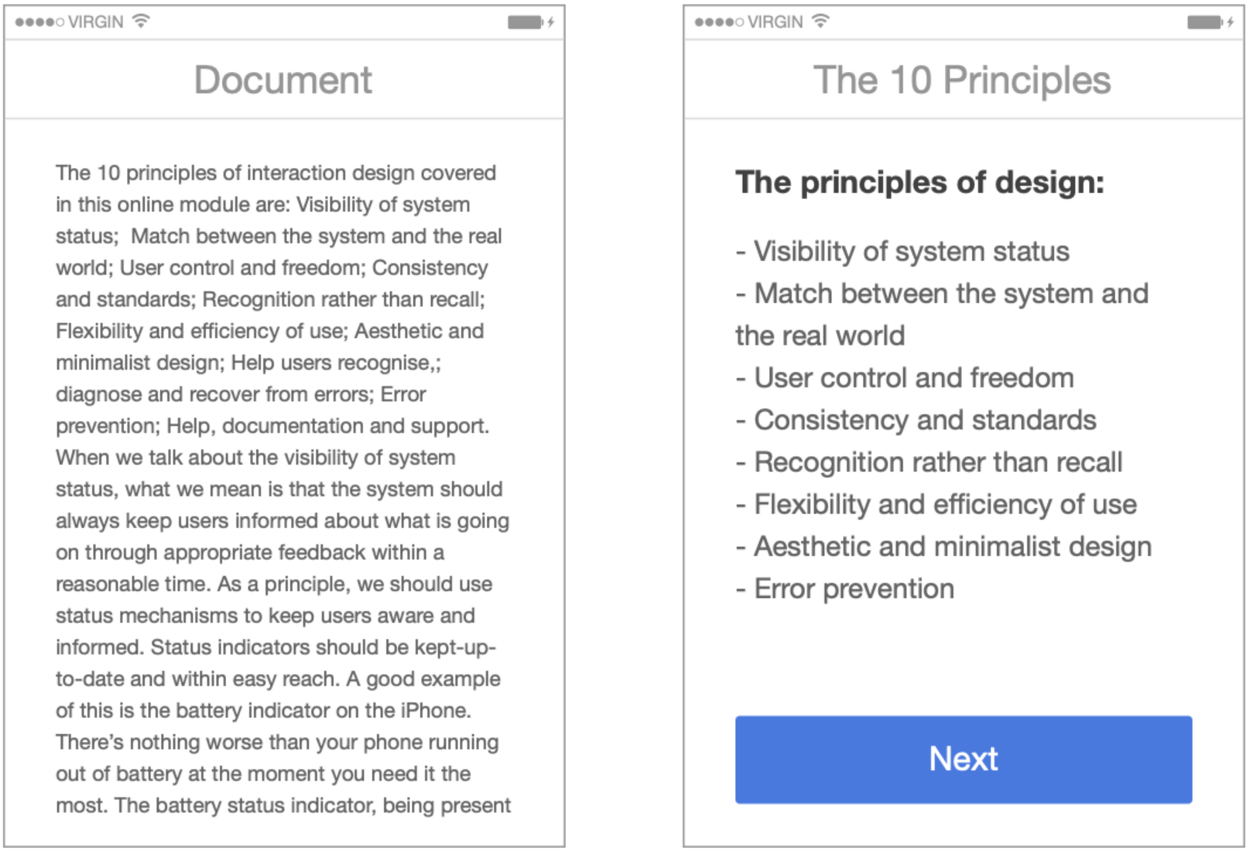
Something unclear in this section? Open an issue.

Aesthetic and minimalist design

Source: Lectures/Interaction-Design/usability-examples/5_aesthetic.md — something unclear? Open an issue, or send a pull request.

Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility.

The image below, designed by Tim Wray shows the difference between aesthetically and in-aesthetically designing the presentation of the ten Nielsen principles.



Counterexamples

The Least Aesthetic Website (Johan)

The website www.arngren.net cotains so much information on the index page, that it is hard for your eyes to find a pattern in the chaos. The content is the opposite of minimalistic and the initial shock is too much for most user. I guess that the average user will last about 2 seconds before leaving the website in horror.

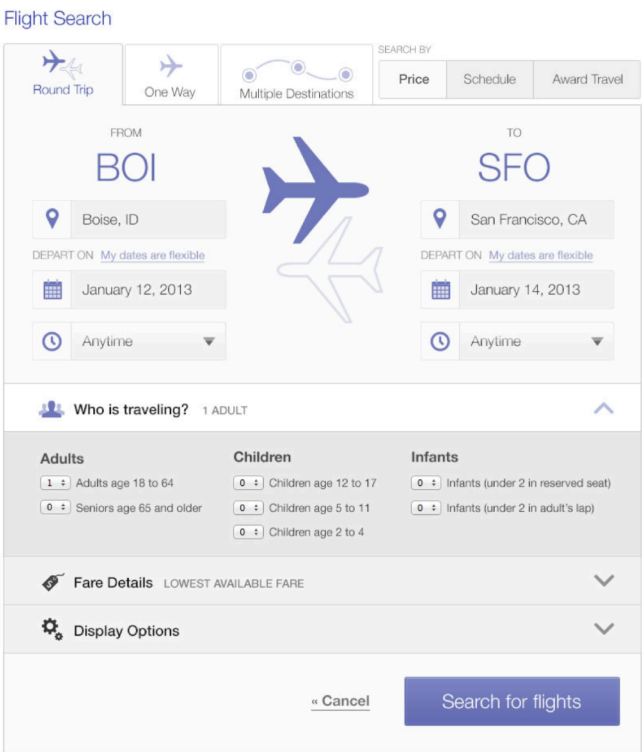
Something unclear in this section? Open an issue.

Recognition rather than recall

Source: Lectures/Interaction-Design/usability-examples/6_recognition_rather_than_recall.md — something unclear? Open an issue, or send a pull request.

Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.

A good example are airline websites which need to keep all the information that the user has selected in a previous step visible, such that the user is confident that they are making the right booking



Source: http://nathanbarry.com/wp-content/uploads/2014/02/redesigned-flight-search.jpg

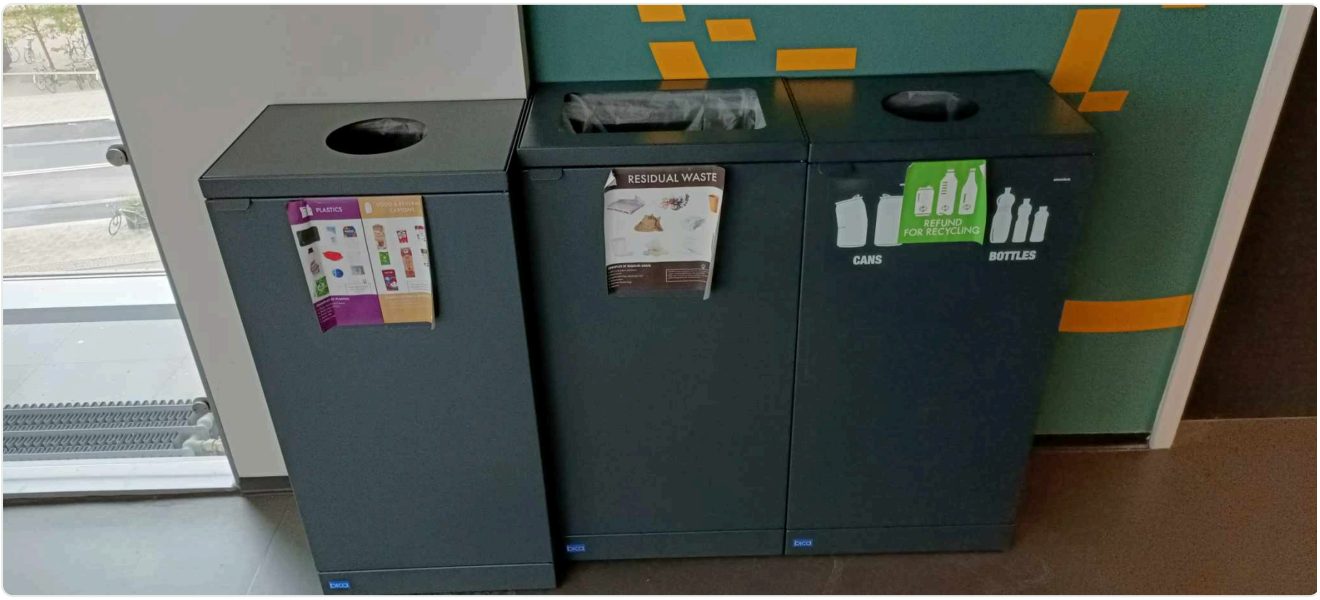
Counterexamples

Unrecognizable Icons

Emma: After very light use, the buttons on my toaster disappeared (they should have made sure the icons were more permanent). For the last couple of years, I have just pressed a random button. I just realized, that the only button that actually means “toast this bread” is the last one. The one that is still visible, because I have never used it. I have been defrosting bread for years - apparently.

Confusing instructions

Mads: When going from one lecture to another in the ITU building, one might want to dispose of some trash. These are the trash cans that are provided. It is however often in a moment when passing by and sometimes in a hurry. I have not yet been able to memorize what goes where, and throwing items out can at times feel like a game of rock, paper, scissor.



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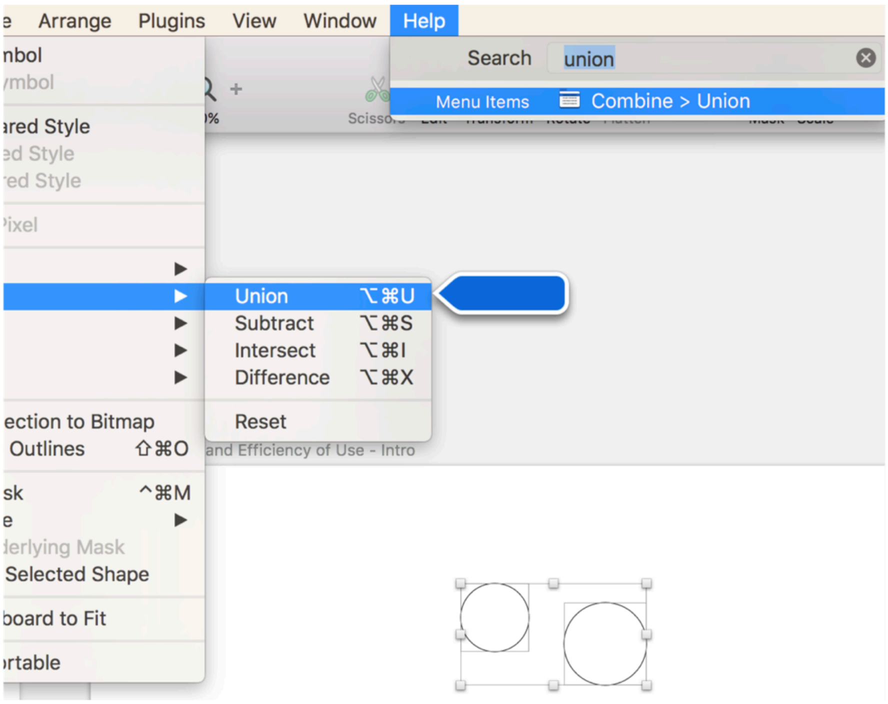
Flexibility and efficiency of use

Source: Lectures/Interaction-Design/usability-examples/7_flexibility_and_efficiency.md — something unclear? Open an issue, or send a pull request.

Accelerators — unseen by the novice user — may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.

Normally, application menus are designed with 6_recognition_rather_than_recall in mind. However, for the expert, there is always a keyboard shortcut that is listed in the menus such that as the user becomes more and more of an expert, they can stop searching in the menus, and start using the shortcuts.

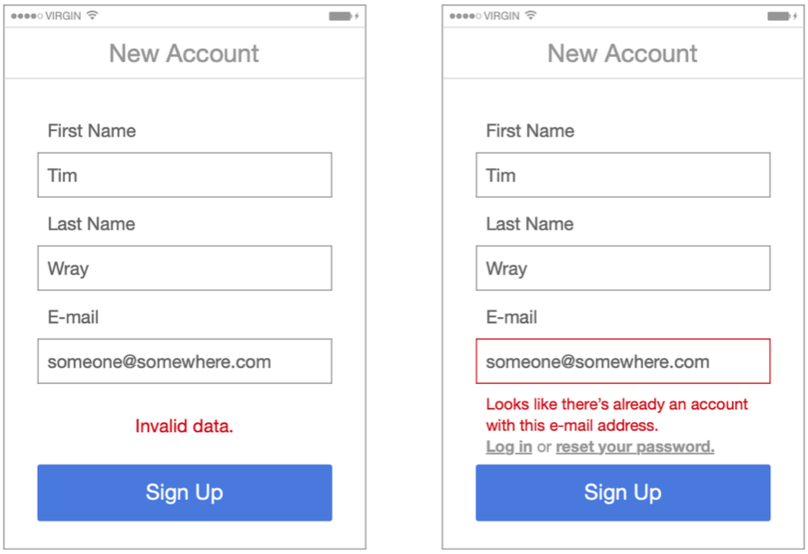


Help users recognize, diagnose, and recover from errors

Source: Lectures/Interaction-Design/usability-examples/8_help_users_recover_from_errors.md — something unclear? Open an issue, or send a pull request.

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

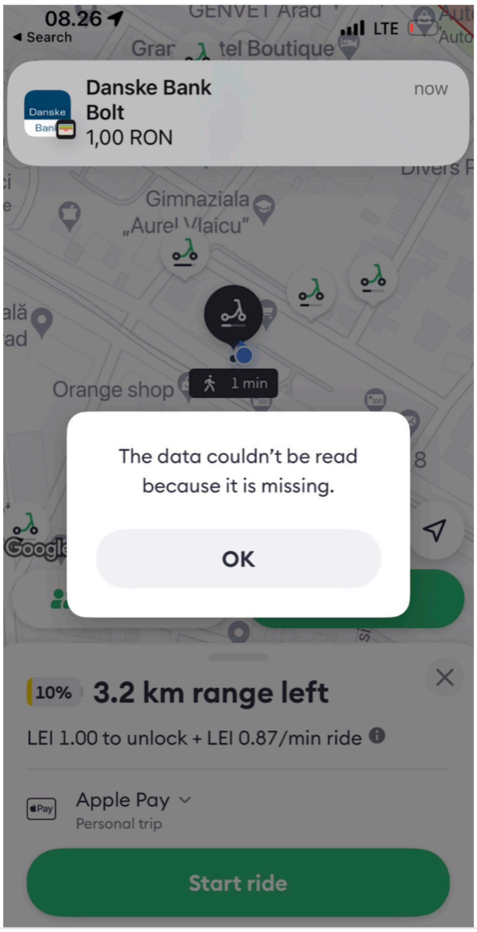
In the example below, Tim Wray shows two possible error messages that could be displayed in a creation dialog for the situation when the user attempts to create an account that already exists.



Counterexamples

Uninformative and Tautological

Mircea: In the summer of 2023, I booked a scooter in Arad, my hometown in Romania using the Bolt app. After I agreed to make a payment of 1RON (approx. 0.25EUR) to prove that my card is valid, and after I saw the notification from the bank that the payment was done, the app showed probably the most useless error message in the history of error messages.



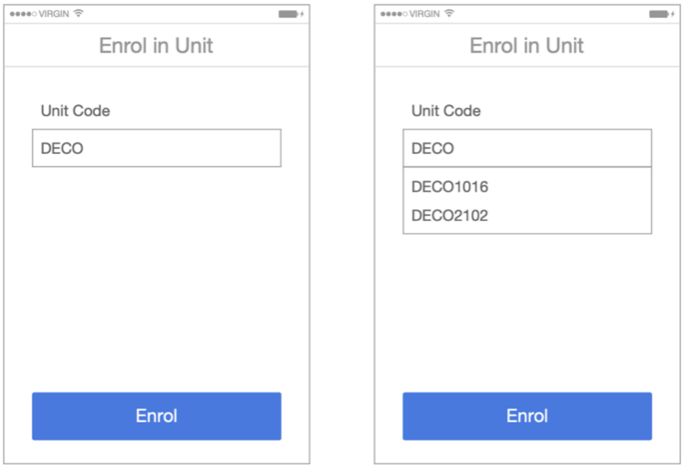
Something unclear in this section? Open an issue.

Error prevention

Source: Lectures/Interaction-Design/usability-examples/9_error_prevention.md — something unclear? Open an issue, or send a pull request.

Even better than good error messages is a careful design that prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.

A possible example is preventing the user from typing errors by providing predefined choices that they can select from.



Help and documentation

Source: Lectures/Interaction-Design/usability-examples/a_documentation.md — something unclear? Open an issue, or send a pull request.

Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.